

Contact

sejalsdesai30@gmail.com

www.linkedin.com/in/sejalsdesai
(LinkedIn)

sejalsdesai30.wixsite.com/sejaldesai (Portfolio)

Top Skills

Visual Basic .NET (VB.NET)

bioreails

One Lab +

Languages

Gujarati (Professional Working)

Punjabi (Limited Working)

English (Native or Bilingual)

Hindi (Native or Bilingual)

Marathi (Native or Bilingual)

Certifications

UX Design: 5 Creating Scenarios and Storyboards

Honors-Awards

Best Publicity Award

Event Head

Star Performer

Value Award

Sejal Desai

Automation & Programming Specialist at Eurofins || Frontend Developer with expertise in UI/UX Designing and Digital marketing United States

Summary

I hold a Master of Science in Information Systems from Northeastern University and a Bachelor of Engineering in Electronics from the University of Mumbai. My academic background combined with hands-on experience has shaped my expertise in web development, automation, and programming.

Currently, I am an Automation and Programming Specialist at Eurofins, where I develop and maintain robotic automation protocols using advanced lab automation tools such as Hamilton Liquid Handlers, Agilent Bravo, and Andrew Pipetting Robots. I streamline workflows with software platforms like VENUS, VBWorks, Green Button Go, and One Lab, enhancing lab productivity by over 30% and reducing equipment downtime by 15%. My role involves cross-department collaboration and thorough documentation to ensure quality compliance and operational consistency.

Previously, as a Web Developer Intern at Zoetis, I designed and launched a responsive inventory tracking system with ReactJS, Tailwind CSS, and TypeScript, improving operational transparency and user experience. I integrated frontend interfaces with backend SQL Server databases through RESTful APIs, applying Agile methodologies to accelerate delivery and foster team collaboration.

I also have experience as a Front-End Web Developer at TCR Innovation, where I developed responsive client-side web applications using HTML, CSS, JavaScript, and jQuery, focusing on usability and cross-browser compatibility.

Continuously advancing my skills, I am proficient in multiple programming languages (Python, Java, C/C++, Swift), web technologies, database systems, and design tools like Figma and Adobe XD. My projects include building e-commerce booking apps

and university application systems that leverage APIs and data parsing to improve functionality and data accuracy.

I am passionate about contributing to innovative projects that combine automation, web development, and user-centric design, aiming to drive efficiency and impactful solutions in technology-driven environments.

Experience

Eurofins

Automation and programming specialist

November 2024 - Present (1 year 6 months)

Michigan, United States

- Spearheaded a cell line development project on Hamilton 2, utilizing Twister channels to automate cap removal and re-capping of 50 mL conical tubes, enabling consistent and repeatable high-throughput sample preparation workflows.
- Engineered a Python-based Hamilton method file extraction tool integrated with an AI model to parse, analyze, and document instrument code logic and execution flow, reducing method review time and accelerating onboarding.
- Optimized Andrew+ pipetting robot experiment runtime by 75% (60 min → 15 min) through method redesign, deck layout optimization, and parallelization of liquid-handling operations, increasing throughput and reproducibility.
- Developed Excel-driven input templates for the glycan assay on Andrew+, streamlining parameter entry, volume mapping, and plate configuration to reduce setup time and minimize manual entry errors.
- Enhanced existing lab automation applications using VB.NET, improving functionality, stability, and usability of tools across active production workflows.
- Designed and validated Hamilton VENUS protocols for the parasitology group, including a flea method requiring precise 2 µL dispense accuracy.
- Collaborated cross-functionally with scientists, QA, and lab operations to align automation protocols with compliance standards, improving assay reproducibility and consistency.

Zoetis

Web Developer

June 2024 - November 2024 (6 months)

Kalamazoo, MI

- Architected and developed a full-stack inventory management solution using React 18, .NET Core 8, and SQL Server to centralize tracking of lab consumables across multiple locations
- Implemented secure JWT authentication and role-based authorization (Admin/User) with RESTful API endpoints for CRUD operations on inventory and order management
- Built responsive UI with Styled Components and React Hooks, featuring dark/light theme toggle, real-time auto-refresh, and mobilefirst design principles
- Designed and optimized SQL Server database schema with Entity Framework Core, including indexed columns and parameterized queries to prevent SQL injection
- Engineered comprehensive error handling and logging system with custom middleware for request validation, exception tracking, and graceful degradation, including automatic session recovery, timeout handling, and user-friendly error messages to ensure 99.5% uptime and seamless user experience across network failures
- Reduced inventory lookup time by 70% and improved order approval workflow efficiency through streamlined admin dashboard and real-time status updates

Indian Society for Technical Education

Secretary General

June 2021 - May 2022 (1 year)

Mumbai

TCR Innovation

Frontend Developer

September 2021 - February 2022 (6 months)

Maharashtra, India

Utilized HTML, CSS, Bootstrap, and JavaScript to develop a responsive and user-friendly website for a pet food company.

Integrated MongoDB to store and manage pet food products and customer data.

Conducted testing and debugging to ensure optimal functionality of the website.

Achieved a visually appealing and highly functional website that met client needs and exceeded expectations.

ILP Overseas | International Learning Planners

Marketing Intern

November 2021 - December 2021 (2 months)

India

worked as an inter was responsible for the marketing things that were carried out by the organisation

CODEOS

Technical Writer

September 2021 - October 2021 (2 months)

Maharashtra, India

worked on technical writing and blog writing

IFORTIS WORLDWIDE™

Sales And Marketing Intern

August 2021 - September 2021 (2 months)

India

worked on marketing strategies and also on social media marketing

Promkraft India Pvt Ltd

Digital Marketing Intern

January 2021 - April 2021 (4 months)

Maharashtra, India

worked on digital platform and social media training

Education

Northeastern University

Masters of Science, Information systems · (September 2022 - May 2024)

Ramrao Adik Institute of Technology

Bachelor's of engineering, Electronics · (August 2018 - May 2022)

S v joshi junior college

Computer Science · (August 2016 - May 2018)

Valinkanni english school dombivli